

Lecture 17 Conservative Vector Fields

Recap: $\vec{F}(x, y, z) = F_x(x, y, z)\vec{i} + F_y(x, y, z)\vec{j} + F_z(x, y, z)\vec{k}$

$$C: \vec{r}(t)$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F} \cdot \vec{r}'(t) dt = \int_C F_x dx + F_y dy + F_z dz.$$

Gradient: $\nabla f = \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j} + \frac{\partial f}{\partial z}\vec{k}.$

Def (Conservative Vector Fields) A vector field $\vec{F}$ is called conservative if and only if it is in the form of $\vec{F} = \nabla f$ where f is a scalar function. f is called a potential function of $\vec{F}$.

Example. Consider the vector field

$$\vec{F}(x, y, z) = (2x+y)\vec{i} + (x+z^3)\vec{j} + (3yz^2+1)\vec{k}$$

Determine whether or not $\vec{F}$ is conservative. If so, find its potential function f .

Sol. $\vec{F} = \nabla f$

$$\begin{cases} \frac{\partial f}{\partial x} = 2x+y & (1) \\ \frac{\partial f}{\partial y} = x+z^3 & (2) \\ \frac{\partial f}{\partial z} = 3yz^2+1 & (3) \end{cases}$$

$$f(x, y, z) = x^2 + yx + g(y, z)$$

$$\frac{\partial f}{\partial y} = x + \frac{\partial g}{\partial y} = x + z^3$$

$$\frac{\partial g}{\partial y} = z^3$$

$$g(y, z) = yz^3 + h(z)$$

$$f(x, y, z) = x^2 + xy + yz^3 + h(z)$$

$$\frac{\partial f}{\partial z} = 3yz^2 + \frac{\partial h}{\partial z} = 3yz^2 + 1$$

$$\frac{\partial h}{\partial z} = 1 \Rightarrow h(z) = z + C$$

$$f(x, y, z) = x^2 + xy + yz^2 + z + C \Rightarrow \vec{F} \text{ is conservative}$$

Example. Let $\vec{F}(x, y) = -y\vec{i} + x\vec{j}$. Determine whether or not $\vec{F}$ is conservative.

If so, find its potential function f .

$$\text{Sol. } \begin{cases} \frac{\partial f}{\partial x} = -y & \text{①} \\ \frac{\partial f}{\partial y} = x & \text{②} \end{cases}$$

$$\text{From ①: } f(x, y) = -xy + g(y)$$

$$\frac{\partial f}{\partial y} = -x + \frac{\partial g}{\partial y} = x$$

$$\frac{\partial g}{\partial y} = 2x \quad \text{not possible!} \Rightarrow \vec{F} \text{ is not conservative.}$$

Theorem. Given a conservative vector field $\vec{F} = \nabla f$, then along any path C connecting from point $P_0(x_0, y_0, z_0)$ to point $P_1(x_1, y_1, z_1)$, the line integral is given by $\int_C \vec{F} \cdot d\vec{r} = f(x_1, y_1, z_1) - f(x_0, y_0, z_0)$.

$$\int_C \nabla f \cdot d\vec{r} = f(x_1, y_1, z_1) - f(x_0, y_0, z_0)$$

Fundamental Theorem of Line Integrals.

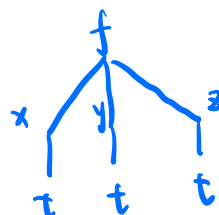
$$\text{Pf. } \vec{F} = \nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

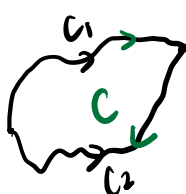
$$C: \vec{r}(t) = \langle x(t), y(t), z(t) \rangle \quad a \leq t \leq b$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \nabla f \cdot \frac{d\vec{r}}{dt} dt$$

$$= \int_a^b \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle \cdot \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle dt$$

$$\begin{aligned}
 &= \int_a^b \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \right) dt \\
 &= \int_a^b \frac{df(\vec{r}(t))}{dt} dt \quad (\text{Chain rule}) \\
 &= f(\vec{r}(b)) - f(\vec{r}(a)) \quad (\text{Fundamental Theorem of Calculus}) \\
 &= f(x_1, y_1, z_1) - f(x_0, y_0, z_0). \quad \square
 \end{aligned}$$





$$\vec{F} \text{ is conservative} \Rightarrow \int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_2} \vec{F} \cdot d\vec{r}$$

$$\oint_C \vec{F} \cdot d\vec{r} = \int_{C_1} \vec{F} \cdot d\vec{r} - \int_{C_2} \vec{F} \cdot d\vec{r} = 0$$

Example. Consider the vector field $\vec{F}(x, y, z) = (2x+y)\vec{i} + (x+z^3)\vec{j} + (3yz^2+1)\vec{k}$, and the path C given by $\vec{r}(t) = (e^t \sin^{2012} t)\vec{i} + (\cos^{2013} t)\vec{j} + \frac{t}{\pi}\vec{k}$ $0 \leq t \leq 2\pi$. Find the line integral $\int_C \vec{F} \cdot d\vec{r}$.

Sol. $\vec{F} = \nabla f$ where $f(x, y, z) = x^2 + xy + yz^3 + ztC$

$$\vec{r}(0) = \langle 0, 1, 0 \rangle$$

$$\vec{r}(2\pi) = \langle 0, 1, 2 \rangle$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \nabla f \cdot d\vec{r} = f(0, 1, 2) - f(0, 1, 0) = (10 + C) - C = 10.$$

Alternatively. L is the straight path connecting $(0, 1, 0)$ and $(0, 1, 2)$

$$\vec{r}_L(t) = \langle 0, 1, 0 \rangle + t(\langle 0, 1, 2 \rangle - \langle 0, 1, 0 \rangle) = \langle 0, 1, 2t \rangle \quad 0 \leq t \leq 1$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_L \vec{F} \cdot d\vec{r}$$

$$= \int_0^1 \langle 2x+y, x+z^3, 3yz^2+1 \rangle \cdot \langle 0, 0, 2 \rangle dt$$

$$= \int_0^1 \langle 1, 8t^3, 12t^2+1 \rangle \cdot \langle 0, 0, 2 \rangle dt$$

$$= \int_0^1 (24t^2+2) dt = (8t^3+2t) \Big|_0^1 = 10.$$

Curl Test.

Def. (Curl of vector field). Given a vector field $\vec{F} = F_x \vec{i} + F_y \vec{j} + F_z \vec{k}$, the curl of $\vec{F}$, denoted by $\text{curl}(\vec{F})$ or $\nabla \times \vec{F}$, is defined as.

$$\begin{aligned}\nabla \times \vec{F} &= \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \times (F_x \vec{i} + F_y \vec{j} + F_z \vec{k}) \\ &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} \\ &= \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \vec{i} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \vec{j} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \vec{k}.\end{aligned}$$

$$\nabla := \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \quad \nabla f = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$$

$$\vec{F}(x,y) = F_x(x,y) \vec{i} + F_y(x,y) \vec{j} + 0 \vec{k}, \quad \nabla \times \vec{F} = \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \vec{k}$$

Example. Find of the curl of $\vec{F} = (2x+y) \vec{i} + (x+z^3) \vec{j} + (3yz^2+1) \vec{k}$.

$$\begin{aligned}\text{Sol. } \nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x+y & x+z^3 & 3yz^2+1 \end{vmatrix} \\ &= \left(\frac{\partial}{\partial y} (3yz^2+1) - \frac{\partial}{\partial z} (x+z^3) \right) \vec{i} + \left(\frac{\partial}{\partial z} (2x+y) - \frac{\partial}{\partial x} (3yz^2+1) \right) \vec{j} \\ &\quad + \left(\frac{\partial}{\partial x} (x+z^3) - \frac{\partial}{\partial y} (2x+y) \right) \vec{k} \\ &= (3z^2 - 3z^2) \vec{i} + (0 - 0) \vec{j} + (1 - 1) \vec{k} \\ &= \vec{0}.\end{aligned}$$